DO NOW — RECALL

Von Neumann and Harvard architectures

Name: _____ Class: _____ Date: _____

1. What is the ideal completion cycle for four independent instructions in a three-stage pipeline?

2. What does MDR contain during an instruction fetch?

3. What happens in the Decode stage?

4. Calculate Sum and Carry-out for a full adder with A=1, B=0, Carry-in=1. Explain in binary.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Cycle 6 (=4+3−1).

2. Model answer

The instruction being transferred from memory.

3. Model answer

The CU identifies and interprets the opcode/operands.

4. Model answer

1+0+1=2 decimal = 10_2, so Sum=0 and Carry-out=1.

Vocabulary — know these terms

Von Neumann — Architecture with shared memory space for instructions and data.

Harvard — Architecture with separate program and data storage paths.

bottleneck — Limit caused by a shared resource or transfer path.

instruction cache — Fast cache storing instructions.

hybrid architecture — Combines features of different architectural approaches.

NEW LEARNING

Read, annotate and explain

Von Neumann organisation

Instructions and data share the same address space and a pathway; access can compete for bandwidth.

Harvard organisation

Instruction and data memories/address spaces are separate, allowing independent access paths.

Recognise contemporary designs

Many processors have aspects of both, such as a unified main memory and separate instruction/data caches (modified Harvard).

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A CPU needs to fetch an instruction while reading its operand. Explain possible contention in a Von Neumann design.

Predict first

The instruction fetch and data read may both need the same memory transfer pathway.

Explain and check

The requests may have to be scheduled rather than served simultaneously; this is related to the Von Neumann bottleneck.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State where instructions and data are stored in Von Neumann and Harvard architectures.

Question 2

A CPU frequently fetches an instruction while reading data. Explain a potential Von Neumann limitation and Harvard advantage.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A modern CPU uses unified main memory but separate L1 instruction and data caches. Explain why it cannot be described as a purely classical Harvard or purely classical Von Neumann design.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Every modern processor is exclusively a pure Von Neumann or pure Harvard machine.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.